

Dr. Venkata Sai Saketh Muddu

✉ venkata.saketh@icts.res.in

🌐 <https://inspirehep.net/authors/1921048>



Employment History

December 2025 -	📌 National Postdoctoral fellow (ANRF, PDF/2025/004764) International center for theoretical sciences (ICTS-TIFR), Bengaluru, India.
August 2025 - December 2025	📌 Postdoctoral fellow International center for theoretical sciences (ICTS-TIFR), Bengaluru, India.
June 2025 - August 2025	📌 Postdoctoral research scholar Max Planck institute for Gravitational Physics, Potsdam, Germany
Dec 2020 - May 2025	📌 PhD research scholar Max Planck institute for Gravitational Physics, Potsdam, Germany
Aug - Dec 2020	📌 Graduate teaching assistant for course PHYS102 - Physics of Sound and Music taught by Dr. Matt Severson at University of Maryland (College Park), MD, USA . I helped students solve homework problems and graded answer sheets.
Jan - May 2020	📌 Graduate teaching assistant for course Physics 161 - General Physics : Mechanics and Particle dynamics taught by Dr. Hailu Gebremariam at University of Maryland (College Park), MD, USA . I helped students solve homework problems and graded answer sheets.
Aug - Dec 2019	📌 Graduate teaching assistant for course Physics 276 - Experimental Physics II: Electricity and Magnetism taught by Dr. Sarah Eno at University of Maryland (College Park), MD, USA . I conducted labs, and graded examinations.

Education

Dec 2020 - May 2025	📌 PHD, Max Planck institute for Gravitational Physics, Potsdam, Germany (Albert Einstein Institute, Potsdam) Research advisor : Dr. Alessandra Buonanno, Director of Astrophysical and cosmological relativity division, Max Planck Institute for Gravitational Physics, Potsdam Dissertation title : PROBING THE NATURE OF COMPACT OBJECTS: SCATTERING, TIDES AND QUASINORMAL MODES PhD Research Topics : • Dynamical tides in Neutron stars and black holes (2407.08327 [gr-qc], Phys.Rev.D 107 (2023) 8, 084006, Phys.Rev.D 109 (2024) 6, 064058) • Quasinormal mode computation and their dependence on compact-object properties. (Phys.Rev.D 110 (2024) 8, 084038, Arxiv:2502.17488) • Spin-induced multipole moments in compact objects. (Phys.Rev.D 106 (2022) 12, 124026) • Classical two body scattering. (Phys.Rev.D 106 (2022) 12, 12402)
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Education (continued)

- Aug 2019 - May 2025  **PHD**, University of Maryland (College Park), Maryland, USA
Academic advisor : Dr. Kaustubh Agashe, Dept. of Physics, University of Maryland (College Park), MD, USA
Academic coursework :
- PHYS624 Advanced Quantum mechanics (A+)
 - PHYS675 Relativity, gravity and cosmology (A+)
 - PHYS759 Seminars on elementary particle physics + QFT (S)
 - PHYS752 Elementary particle physics II theory (S)
 - PHYS759 Seminars on elementary particle physics + QFT (S)
 - PHYS851 Advanced Quantum field theory (S)
- GPA** = 4.00
- Aug 2015 - Jul 2019  **Bachelors of Science in Physics**, Indian institute of Technology, Kanpur, India
CPI (Cumulative performance index) = 9.6/10.0, graduated with distinction.

Skills

- Coding  C++, Python, Julia, $\text{\LaTeX}$, Mathematica,
Misc.  Strong analytical skills, Academic research, teaching, training, $\text{\LaTeX}$ typesetting and publishing.
Languages  English, Telugu, Hindi.

Relevant experience and awards

Conferences attended and talks given

- 13th–24th July 2026  Served as a **tutor** for the graduate-level courses **Effective Field Theory Approach to Post-Newtonian Gravity**, taught by Jan Steinhoff, and **Applications of Scattering Amplitudes to Compact Binary Inspirals**, taught by Donal O’Connell, at the **ICTS Summer School on Gravitational-Wave Astronomy**, ICTS-TIFR, Bengaluru, India.
- 29th June 2026  Gave a talk titled **Dynamical tidal response in neutron stars** (based on work in Arxiv 2606.14405 [gr-qc]) as an **invited seminar speaker** at **Indian association for cultivation of sciences (IACS)**, Kolkata, India.
- 19th May 2025  Oral defense of dissertation titled **PROBING THE NATURE OF COMPACT OBJECTS: SCATTERING, TIDES AND QUASINORMAL MODES** based on research carried out during the PhD at **Max Planck Institute for Gravitational Physics, Potsdam**.
- 6th March 2025  Gave a talk titled **Tidal response of compact objects from relativistic Raman scattering** (based on work in Phys.Rev.D 107 (2023) 8, 084006, Phys.Rev.D 109 (2024) 6, 064058) as an **invited seminar speaker** for the **Astrorel seminar** at **International institute for theoretical sciences (ICTS-TIFR)**, Bangalore, India.

Relevant experience and awards (continued)

- 29th November 2024 ■ Gave a talk titled **Dynamical tidal response in relativistic Compact objects** (based on work in Phys.Rev.D 107 (2023) 8, 084006, Phys.Rev.D 109 (2024) 6, 064058) as an **invited seminar speaker** for the **Strong Seminar** at **Neils Bohr Institute, University of Copenhagen**.
- 29th July 2024 ■ Gave a talk presenting **Investigating tidal heating in neutron stars via gravitational Raman scattering** (2407.08327 [gr-qc]) as an **external speaker** at **Extreme matter call, LIGO**.
- 6th Mar 2024 ■ Gave a talk on **Dynamical tidal response of Kerr black holes using effective worldline theory** (Phys.Rev.D 107 (2023) 8, 084006, Phys.Rev.D 109 (2024) 6, 064058) as an **invited seminar speaker** at **Inter-University centre for Astronomy and Astrophysics (IUCAA), Pune, India**
- 31st Jan 2024 ■ Gave a talk on **Dynamical tidal response of Kerr black holes using effective worldline theory** (Phys.Rev.D 107 (2023) 8, 084006, Phys.Rev.D 109 (2024) 6, 064058) as an **invited seminar speaker** at **Max Planck institute for Gravitational Physics, Hannover, Germany**
- 8th Dec 2023 ■ Gave a talk presenting **Membrane paradigm for slowly spinning compact objects** (Phys.Rev.D 110 (2024) 8, 084038) in **10th international conference on gravitationa and cosmology : New horizons and singularities in Gravity (ICGC 2023)** at **Indian institute of technology, Guwahati, India**
- July 2023 ■ Attended "**Amplitudes to gravitational waves**" conference at **Nordita, Stockholm, Sweden**
- 5th Dec 2022 ■ Gave a talk presenting **Modeling horizon absorption in spinning binary black holes using effective worldline theory** (Phys.Rev.D 110 (2024) 8, 084038) at **GRAMPA seminar** via zoom.
- Dec 2022 ■ Attended **QCD meets gravity 2022** at **University of Zurich, Switzedland**
- April - May 2022 ■ Attended **KITP gwaves22** conference at **University of California, Santa Barbara, USA**
- 22nd March 2022 ■ Gave a talk presenting **Scattering gravitational waves off effective spinning black holes** (Phys.Rev.D 106 (2022) 12, 124026) in **DPG Spring Meeting "Heidelberg22"** via Zoom

Awards and Achievements

- 2026 ■ **National postdoctoral fellowship (ANRF)**, The ANRF-National Post Doctoral Fellowship (N-PDF) is aimed at identifying motivated young researchers and providing them with support for doing research in frontier areas of science and engineering.
- 2025 ■ **Misner award**, established in honor of Professor Charles Misner, to recognize outstanding thesis work in gravitation and cosmology.
- 2020-21 ■ **Richard and Anna Iskraut scholarship**, for the first-year graduate student with highest score in the classical portion of the Qualifying Examination, at the University of Maryland (College Park), MD, USA
- July 2019 ■ **Obtained BS degree in physics with distinction from Indian institute of technology Kanpur**
- May 2015 ■ **Qualified JEE advanced 2015 (All India Rank : 1354)**
- 2014-15 ■ **Qualified Indian national physics olympiad 2015 and attended the training camp for international physics olympiad**

Relevant experience and awards (continued)

- **Qualified Kishore Vaigyanik Protsahan Yojana (KVPY) exam and received scholarship to support my undergraduate studies.**

References

- Dr. Alessandra Buonanno** a) Director of Astrophysical and Cosmological Relativity division, at Max Planck institute for Gravitational Physics, Am Mühlenberg 1, 14476 Potsdam, Germany. b) Research professor at the University of Maryland (College Park), MD, USA. c) Honorary Professor of Physics, Humboldt University, Berlin d) Honorary Professor of Physics, Potsdam University, Potsdam.
Email : alessandra.buonanno@aei.mpg.de
- Dr. Nils Anderson** Professor of Applied Mathematics at the University of Southampton, United Kingdom.
Email : N.A.Andersson@southampton.ac.uk
- Dr. Jan Steinhoff** a) Group Leader at Max Planck institute for Gravitational Physics, Am Mühlenberg 1, 14476 Potsdam, Germany. b) Professor at Humboldt University, Berlin.
Email : jan.steinhoff@aei.mpg.de
- Dr. Julio Parra-Martinez** Senior Professor of Physics at Institut des Hautes Études Scientifiques Université Paris-Saclay.
Email : julio@ihes.fr
- Dr. Tanja Hinderer** a) Associate professor at Utrecht University. b) Co-chair of the Division 'Nuclear Physics' on the Einstein Telescope Observational Science Board.
Email : t.p.hinderer@uu.nl
- Dr. Justin Vines** Senior researcher, Joint position at Max Planck institute for Gravitational Physics, Potsdam and University of California Los Angeles (UCLA), Los Angeles, CA 90095, USA.
Email : vines@physics.ucla.edu, justin.elliott.vines@gmail.com
- Dr. Debarati Chatterjee** Associate professor at Inter-University Centre for Astronomy and Astrophysics, Ganeshkhind, Pune - 411007, Maharashtra, India.
Email : debarati@iucaa.in
- Dr. Parameshwaran Ajith** Professor at International Centre for Theoretical Studies (ICTS-TIFR), Survey No. 151, Shivakote, Hesaraghatta, Uttarahalli Hobli, Bengaluru, Karnataka 560089, India.
Email : ajith@icts.res.in
- Dr. Pankaj Jain** Professor, Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE), Indian Institute of Technology Kanpur (IIT-K).
Email : pkjain@iitk.ac.in

References (continued)

Research Publications

Journal Articles

- 1 S. Chakraborty, M. V. S. Saketh, T. Hinderer, and J. Steinhoff, “Dynamical tidal Love numbers of black holes under generic perturbations: Connecting black hole perturbation theory with effective field theory,” *Phys. Rev. D*, vol. 114, no. 6, p. 064 011, 2026.  DOI: 10.1103/lfq4-49k1. arXiv: 2605.00693 [gr-qc].
- 2 M. V. S. Saketh, R. Ghosh, and A. Mishra, “Strong-field gravitational-wave lensing in the Kerr background,” *Phys. Rev. D*, vol. 113, no. 8, p. 084 056, 2026.  DOI: 10.1103/9bxg-sqn5. arXiv: 2511.23110 [gr-qc].
- 3 M. V. S. Saketh, S. Ghosh, and N. Andersson, “Dynamical tidal response of neutron stars via scattering amplitudes,” *Phys. Rev. D*, vol. 114, no. 6, p. 064 008, 2026.  DOI: 10.1103/gvv9-cnmx. arXiv: 2606.14405 [gr-qc].
- 4 H. O. Silva, J.-W. Kim, and M. V. S. Saketh, “Kerr-Newman quasinormal modes and Seiberg-Witten theory,” Feb. 2025. arXiv: 2502.17488 [gr-qc].
- 5 M. V. S. Saketh and E. Maggio, “Quasinormal modes of slowly-spinning horizonless compact objects,” *Phys. Rev. D*, vol. 110, no. 8, p. 084 038, 2024.  DOI: 10.1103/PhysRevD.110.084038. arXiv: 2406.10070 [gr-qc].
- 6 M. V. S. Saketh, Z. Zhou, S. Ghosh, J. Steinhoff, and D. Chatterjee, “Investigating tidal heating in neutron stars via gravitational raman scattering,” *Phys. Rev. D*, vol. 110, p. 103 001, 10 Nov. 2024.  DOI: 10.1103/PhysRevD.110.103001.
- 7 M. V. S. Saketh, Z. Zhou, and M. M. Ivanov, “Dynamical tidal response of Kerr black holes from scattering amplitudes,” *Phys. Rev. D*, vol. 109, no. 6, p. 064 058, 2024.  DOI: 10.1103/PhysRevD.109.064058. arXiv: 2307.10391 [hep-th].
- 8 M. V. S. Saketh, J. Steinhoff, J. Vines, and A. Buonanno, “Modeling horizon absorption in spinning binary black holes using effective worldline theory,” *Phys. Rev. D*, vol. 107, no. 8, p. 084 006, 2023.  DOI: 10.1103/PhysRevD.107.084006. arXiv: 2212.13095 [gr-qc].
- 9 A. Vijaykumar, M. V. S. Saketh, S. Kumar, P. Ajith, and T. R. Choudhury, “Probing the large scale structure using gravitational-wave observations of binary black holes,” *Phys. Rev. D*, vol. 108, no. 10, p. 103 017, 2023.  DOI: 10.1103/PhysRevD.108.103017. arXiv: 2005.01111 [astro-ph.CO].
- 10 M. V. S. Saketh and J. Vines, “Scattering of gravitational waves off spinning compact objects with an effective worldline theory,” *Phys. Rev. D*, vol. 106, no. 12, p. 124 026, 2022.  DOI: 10.1103/PhysRevD.106.124026. arXiv: 2208.03170 [gr-qc].
- 11 M. V. S. Saketh, J. Vines, J. Steinhoff, and A. Buonanno, “Conservative and radiative dynamics in classical relativistic scattering and bound systems,” *Phys. Rev. Res.*, vol. 4, no. 1, p. 013 127, 2022.  DOI: 10.1103/PhysRevResearch.4.013127. arXiv: 2109.05994 [gr-qc].
- 12 R. Kothari, M. V. S. Saketh, and P. Jain, “Torsion driven Inflationary Magnetogenesis,” *Phys. Rev. D*, vol. 102, no. 2, p. 024 008, 2020.  DOI: 10.1103/PhysRevD.102.024008. arXiv: 1806.02505 [astro-ph.CO].